

Leakage-Free Operational Earthquake Forecasting: A Multi-Region CSEP Testbed with Pre-Registered Negative Results and the Horizon-Dependent Value of Geodetic Context

Felipe Santib  n  z-Leal , *Independent Researcher*

Preprint · v2.2, 18 September 2026 · Licence CC BY 4.0

CAOS open-research programme · CAOS_SEISMIC, operational earthquake forecasting

Version DOI 10.5281/zenodo.22822107 · concept DOI (always latest) 10.5281/zenodo.21508362

Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_SEISMIC

Abstract. Operational earthquake forecasting (OEF) issues calibrated conditional probabilities of future seismicity; the Epidemic-Type Aftershock Sequence (ETAS) model is its de-facto benchmark, and under fair, prospective, CSEP-style testing no machine-learning temporal point process has been shown to beat a well-fit ETAS. We describe a complete OEF system and, on it, a leakage-free, multi-region CSEP testbed designed to test four questions empirically. The system fits a regime-tiled space-time ETAS with a full hygiene pipeline (rolling magnitude of completeness, moment-magnitude homogenization, dual-catalog declustering, propagated uncertainty), against a mandatory adaptive smoothed-seismicity Poisson null and a transparent Reasenberg-Jones fallback, calibrated by isotonic regression with an epistemic-plus-aleatory uncertainty triad, and it emits both gridded and catalog-based forecast representations so that over-dispersion is accounted for in scoring. Skill is established only by winning CSEP comparison tests (information gain per earthquake, IGPE, in nats) against both baselines, under a strict forecast clock that guards against five leakage modes, with adoption rules fixed before each run and shuffled-label negative controls. Four findings result. (i) ETAS adds skill over the stationary null only where there is triggering to exploit (Japan +0.072 nats at one day; low-seismicity interiors exactly 0.0), quantifying the high-versus-low-seismicity bias. (ii) A convex log-score-optimal stack of ETAS-family variants earns a significant *global* gain (+0.011 over 751 events) that does not generalize to the canonical active margins, so the pre-registered adoption rule rejects it; a temporally adaptive variant that appears to rescue it is a multiple-comparison artifact. (iii) The binding one-to-seven-day consistency failure is count over-dispersion, not spatial shape: a catalog-based number test passes a window the Poisson number test rejects, and the frozen-intensity forecast systematically under-counts by about 28% because it omits within-window secondary triggering. (iv) The value of a GNSS-strain geodetic context covariate, added to a Hawkes-structured neural point process, is *horizon-dependent*: it does not beat ETAS at the one-to-seven-day operational horizon (mean IGPE -0.053 over eight weekly windows) but beats it robustly at thirty days

(global +0.115 over 2166 events, positive in 9/10 windows and in every high-seismicity region), because the calibrated model deploys a time-flat geodetic background rather than triggering. We conclude that base tiled ETAS is at or near the practical ceiling for mean-rate one-to-seven-day IGPE over global $M \geq 5$, and that a geodetic covariate belongs in a longer, background-dominated outlook.

Index Terms. operational earthquake forecasting, ETAS, epidemic-type aftershock sequence, CSEP, pyCSEP, information gain, prospective evaluation, leakage, over-dispersion, catalog-based testing, neural temporal point process, geodetic strain, GNSS, calibration, reproducibility.

I. INTRODUCTION

EARTHQUAKES cannot be predicted deterministically: whether a small rupture cascades into a great earthquake depends on unmeasurably fine details of the crust [1]. What can be done, and is done operationally, is *forecasting*: issuing a probability strictly in $(0, 1)$ for a region, a magnitude band, and a short horizon. Following the ICEF definition [2], a prediction is a deterministic statement that an event will or will not occur, whereas a forecast reports a probability with uncertainty; short-term probabilities of a large event may vary over orders of magnitude relative to background yet typically remain small in absolute terms (well under a few percent per day). This distinction is not pedantic. The documented harm in the field is *false reassurance*, so every number here is a conditional probability shown next to its long-term baseline, scored prospectively, and never rendered as an alarm.

The de-facto short-term baseline is the Epidemic-Type Aftershock Sequence model [3], [4], a self-exciting Hawkes point process. Operational systems such as the USGS aftershock forecast [5], [6] and OEF-Italy [7] run ETAS-class models as scheduled services; a decade of prospective next-day forecasting over California confirms that no single model dominates, with the short-term-clustering models excelling during sequences and ETAS the consistent generalist [8]. Against this baseline, machine learning has repeatedly promised more and, under fair testing, repeatedly failed to deliver a prospective gain: the EarthquakeNPP benchmark [9] found that none of

F. Santib  n  z-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanezleal@ug.uchile.cl; ORCID: 0000-0002-0150-3246).

Code, configurations, provenance manifests, and the daily and back-analysis artifacts are released under MIT at https://github.com/fsantibanezleal/CAOS_SEISMIC; a static forecast viewer is available at <https://seismic.fasl-work.com>. This is an independent research and education tool; it is not an operational alarm system and must not be used for life-safety decisions.

five modern neural point processes beat ETAS once strict chronological splits and CSEP consistency tests are imposed. Where learned models can help, the gains come from two ETAS gaps, ingesting covariates ETAS cannot easily absorb, and learning spatial anisotropy, not from network depth [10], [11].

This paper makes two kinds of contribution. First, it documents a complete OEF system: the estimator stack (Section III), its calibration and uncertainty, and a leakage-free, multi-region CSEP evaluation protocol (Section IV) with a strict forecast clock, pre-registered adoption rules, and shuffled-label negative controls. Second, on that testbed it reports four findings (Section VI), together with their negative results: the high-versus-low-seismicity bias of ETAS; an ETAS-family stack that gains globally but not regionally; the over-dispersion and secondary-cascade nature of the binding short-horizon consistency failure; and the horizon-dependent value of a GNSS-strain geodetic covariate. Every reported number is reproducible from the released artifacts.

II. DATA

The catalog spine is USGS ComCat (FDSN event service, public domain), homogenized to moment magnitude M_w . Where a per-network station-pair regression to M_w is unavailable, the Scordilis [12] global $m_b \rightarrow M_w$ and $M_s \rightarrow M_w$ orthogonal-regression relations are used as a documented fallback with per-event provenance; this recovers a six-fold larger conditioning catalog (from 1.65×10^4 to 9.7×10^4 events, $M \geq 4.5$, 1990–2026) at the cost of a bounded, quantified b -value inflation that the per-tile fit largely removes (a naive global $b \approx 1.59$ becomes a per-tile production $b \approx 1.34$). The magnitude of completeness M_c and the Gutenberg–Richter b -value are estimated on rolling windows with their uncertainty [13], [14], the Aki–Utsu b carrying its Shi–Bolt standard error [15]. Declustering, where the stationary background requires it, uses stochastic declustering [16] and the nearest-neighbour method [17]; the scored target, however, is the *non-declustering* catalog, because clustering is what we forecast. The geodetic covariate is the GNSS strain rate from the Nevada Geodetic Laboratory MIDAS velocity field [36] (20,168 stations), gridded to a coarse global field; strain is high on active margins (Japan 95, California 154, Chile 64 nstrain/yr) and low in stable interiors (Central US 3.6, Western Europe 5.6). Plate-boundary geometry is PB2002 [37]. Raw catalogs, features, and model weights can be rebuilt from the code and provenance manifests; the compact daily and back-analysis artifacts are archived in the repository.

III. MODELS: THE ESTIMATOR STACK

Every model implements one seam, the Forecaster port `fit→expected_counts`, so the null, the reference, and any challenger are fit on the identical conditioning slice and scored on identical bins.

A. The public target: a bounded exceedance probability

The single number a user sees is the probability of at least one event above a target magnitude M^* in a cell over a horizon $[0, T]$, a non-homogeneous Poisson exceedance probability

$$P(\geq 1 \text{ event} \geq M^*) = 1 - e^{-N_{\geq M^*}}, \quad (1)$$

$$N_{\geq M^*} = \int_0^T \int_A \lambda \Phi(M^*) dx dy dt,$$

where the magnitude term is the Gutenberg–Richter tail fraction, *bounded* by a per-region maximum magnitude $M_{\max}$ so an unbounded tail can never inflate the rare-event probability,

$$\Phi(M^*) = \frac{10^{-b(M^*-M_c)} - 10^{-b(M_{\max}-M_c)}}{1 - 10^{-b(M_{\max}-M_c)}}, \quad (2)$$

for $M_c \leq M^* \leq M_{\max}$. The public formula $P = 1 - e^{-N}$ never changes; only the quality of the conditional intensity λ improves. The full conditional magnitude distribution is forecast (not a single threshold), which is what enables the CSEP magnitude test and keeps the display region-appropriate.

B. The mandatory null: adaptive smoothed seismicity

A stationary Poisson estimate of where earthquakes occur, obtained by smoothing a declustered catalog with the Helmstetter–Kagan–Jackson adaptive power-law kernel [18],

$$\mu(x, y) = \sum_i K_{d_i}(r_i), \quad K_d(r) = C(d) (r^2 + d^2)^{-s}, \quad (3)$$

with bandwidth d_i the distance to event i 's n -th nearest neighbour. It is both the spatial background that seeds ETAS and the mandatory stationary reference any time-dependent model must beat. The exponent s and normalization $C(d)$ are pinned to named reference profiles, never assumed.

C. The primary estimator: space–time ETAS

ETAS is a self-exciting Hawkes process, a stationary background plus the summed, decaying offspring of every past event [4],

$$\lambda(t, x, y | \mathcal{H}_t) = \mu(x, y) + \sum_{i: t_i < t} K e^{\alpha(M_i - M_0)} \left(1 + \frac{t - t_i}{c}\right)^{-p} f(x - x_i, y - y_i | M_i), \quad (4)$$

with Utsu productivity $k(m) = K e^{\alpha(m - M_0)}$ (reference magnitude M_0), the Omori–Utsu kernel $g(t) = \frac{p-1}{c} (1 + t/c)^{-p}$, and the Ogata inverse-power spatial kernel $f(r | M_i) = \frac{q-1}{\pi \zeta^2} (1 + r^2/\zeta^2)^{-q}$, $\zeta = D e^{\gamma(M_i - M_0)}$. Parameters $(K, \alpha, c, p, D, \gamma, q)$ are fit by maximum likelihood on the full un-declustering catalog, maximizing the point-process log-likelihood

$$\ln L = \sum_i \ln \lambda(t_i, x_i, y_i | \mathcal{H}_{t_i}) - \int_0^T \int_A \lambda dx dy dt. \quad (5)$$

Two distinct stability conditions are enforced: finite branching ($\alpha < \beta$, $\beta = b \ln 10$) and subcriticality (branching ratio $n < 1$); a supercritical fit signals a mis-fit and is rejected. Because a single global ETAS over a 10^5 -event catalog is intractable and

physically wrong (subduction and stable interior obey different kinematics), we fit a *regime-tiled* ETAS: 195 ETAS tiles are fit and 265 sparse tiles fall back to their smoothed null, at $M_c = 5.35$, $b = 1.337$.

D. The transparent fallback: Reasenberg–Jones

The most transparent aftershock model [5], [6] conditions on the largest triggering event before issue time, with rate $\lambda(t, M) = 10^{a+b(M_m-M)}(t+c)^{-p}$ (M_m the conditioning mainshock magnitude), distributed over cells by a Wells–Coppersmith rupture-length kernel [19]; its constants are flagged as California-derived and never silently reused for Chilean subduction. It is a human-auditable cross-check, not the primary spatial forecaster.

E. The neural challenger with geodetic context

The stronger model is delivered in two layers; the second is used only if it passes the comparison tests. Layer 1 is the region-refit tiled ETAS with the full hygiene pipeline. Layer 2 is a conditional spatio-temporal neural point process with a Hawkes inductive bias, in the spirit of RECAST [10] and FERN [11]: keep the additive background plus summed-triggering skeleton, replace the fixed kernels with learned maps, and ingest a gridded covariate field, here the GNSS strain-rate channel (the other geophysical channels are zero-filled and no result depends on them). A fit-time scalar anchors the integrated one-day forecast rate to the training rate on the evaluation grid; without it the background head over-predicts the absolute rate by roughly $490\times$ (a failure we diagnose and report in Section VI-D). The challenger enters the public forecast only if it beats ETAS in the prospective CSEP tests and is calibrated; otherwise ETAS remains the operational model.

F. Calibration and uncertainty

The public probability is recalibrated by isotonic regression and validated with a reliability diagram per horizon; an uncalibrated probability is never published. The uncertainty triad (optimistic P10, expected median, pessimistic P90) [20] is an epistemic-plus-aleatory decomposition sourced from ETAS parameter covariance, propagated M_c/b uncertainty, model-selection uncertainty, and *over-dispersion*: because regional seismicity has variance well above its mean, the pessimistic bound is wider than a naive Poisson quantile (negative-binomial behaviour [21]). A pessimistic bound computed as a Poisson quantile systematically under-warns at the tail. In the highest-stakes window, immediately after a large mainshock, the daily job uses an incompleteness-aware likelihood with a time-dependent $M_c(t)$, the failure mode behind OEF-Italy’s documented under-estimation of the 2016–2017 Central Italy sequence [7].

IV. EVALUATION: A LEAKAGE-FREE CSEP TESTBED

A forecast can only be judged by prospective scoring. The scoring framework is CSEP via pyCSEP [22]; using standard tests rather than bespoke metrics lets a reviewer dispute the model rather than the test code.

A. Two representations and over-dispersion

We emit both a gridded-rate forecast (Poisson expected count per space–magnitude cell) and a catalog-based forecast (an ensemble of synthetic catalogs). Regional seismicity is over-dispersed relative to Poisson, so the Poisson grid tests over-reject during sequences; the catalog-based tests [23] relax that assumption and are the correct primary path for a clustered daily forecast, always reported paired with their Poisson counterpart.

B. The forecast clock and five leakage modes

The primary mode is pseudo-prospective [24], driven by a forecast clock (Fig. 1): at each daily issue time t the model is handed only the catalog slice up to t , the forecast is sealed, then the clock advances; the conditioning window is half-open on the left and the target $[t, t+H)$ half-open on the right, so back-to-back horizons tile the timeline without double-counting, and an always-on invariant asserts no leakage before every fit. Five leakage modes are engineered against: temporal leakage (the clock); catalog-revision leakage (an immutable input snapshot per issue makes each forecast byte-reproducible); M_c -inconsistency leakage (one uniform $M_{\min}$ applied to the model and every baseline); region/parameter snooping (pre-registered polygons, horizons, thresholds, grid, and declustering, frozen across the test split); and multiple-testing inflation (every region $\times$ horizon cell reported, including failures). To keep the multi-region back-analysis tractable the ETAS parameters are fit on a weekly cadence and only the triggering conditioning is reconditioned daily, a measurement-preserving change validated to a 0.05% median deviation from full-refit-every-day.

C. Consistency and comparison tests

Consistency (calibration) tests are built on the Poisson joint log-likelihood over space–magnitude bins, $L(\Omega | \Lambda) = \sum_i [-\lambda_i + \omega_i \ln \lambda_i - \ln(\omega_i!)]$: the number (N) test with Poisson-tail quantiles $\delta_1 = 1 - F(N_{\text{obs}} - 1 | N_{\text{fore}})$ and $\delta_2 = F(N_{\text{obs}} | N_{\text{fore}})$, the magnitude (M), spatial (S), likelihood (L) and preferred conditional-likelihood (CL) tests [25], [26]. These are necessary but not sufficient: passing consistency calibrates one model but never establishes skill, because a trivial “aftershocks follow mainshocks” model also passes. Skill is established only by winning a comparison test against both the smoothed-seismicity null and ETAS, measured as information gain per earthquake (IGPE, nats),

$$I_N(A, B) = \frac{1}{N} \sum_{i=1}^N (\ln \lambda_A(k_i) - \ln \lambda_B(k_i)) - \frac{\hat{N}_A - \hat{N}_B}{N}, \quad (6)$$

where k_i is the space–magnitude bin of the i th target event; the gain is tested with the paired t -test $T = I_N/(s/\sqrt{N}) \sim t_{N-1}$ and corroborated by the Wilcoxon W-test [27]. The gain over ETAS measures genuine added skill precisely because both models already reproduce Omori clustering. ROC/AUC is not used as a skill metric: it is invariant to calibration and, on a rare per-cell-per-day task, degenerates into a region classifier (the DeVries trap [28], [29]). Proper scoring rules [30] and

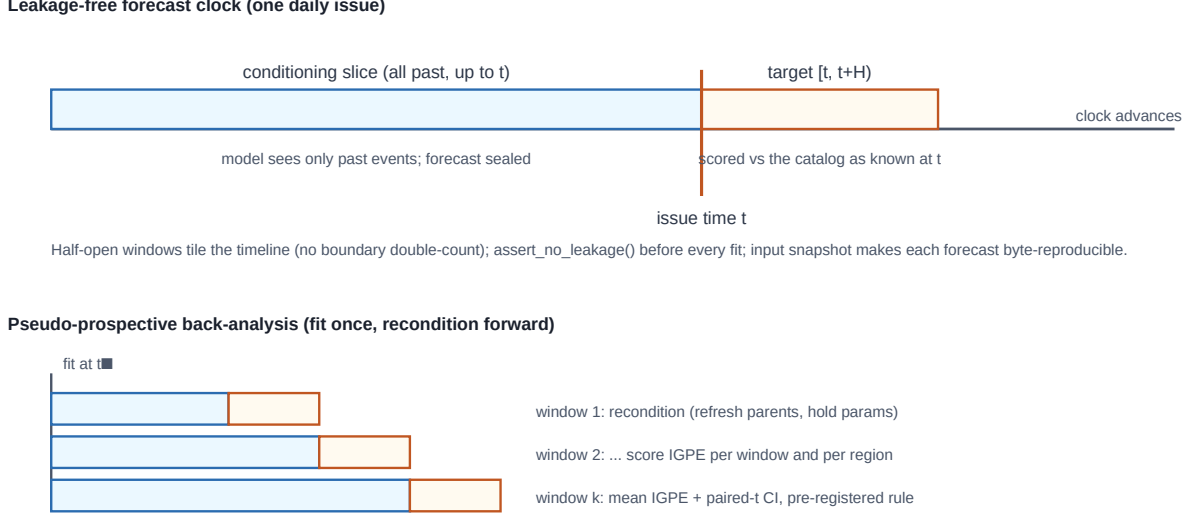


Fig. 1. The leakage-free forecast clock and the pseudo-prospective back-analysis. At each daily issue t the model receives only the conditioning slice up to t , the forecast is sealed, and it is scored against the catalog as known at t ; half-open windows tile the timeline without double-counting. The back-analysis fits once and reconditions forward across weekly windows (refresh parents, hold parameters), scoring IGPE per window and per region under a pre-registered rule.

a Molchan/area-skill view [31] are reported only as communication aids. Adoption rules are fixed before each run (pre-registration), and a shuffled-label negative control guards against over-fitting.

V. RELATED WORK: MACHINE LEARNING VERSUS ETAS

As of writing, no machine-learning model has been shown to reliably beat a well-fit ETAS for short-term forecasting under fair, prospective, CSEP-style testing; our protocol tests that requirement and does not assume that machine learning can never help. The decisive evidence is EarthquakeNPP [9]: five modern neural point processes on California 1971–2021 with strict chronological splits and CSEP consistency tests, none of which outperformed ETAS, and specifically the best neural model reached only about 68.6% on the spatial test where forecasting value lives, versus 92.0% for ETAS. The methodological fix was repairing a data-leakage flaw in earlier neural work (non-chronological splits inflate metrics via triggering); once temporal splits and the large sequences are restored the apparent advantage evaporates. The published gains come from ETAS gaps, not from depth: RECAST improves on temporal ETAS only for large training catalogs [10], and FERN’s reported information-gain improvement came mostly from ingesting sub- M_c events and learning fault-aligned anisotropy, with the authors’ own caveats (not CSEP-tested, no uncertainty quantification), which our protocol treats as requirements [11]. The geodetic input we study is grounded in long-term global rate models: GEAR1 blends smoothed seismicity with a strain-rate model [34], and prospective global hybrids find strain adds skill globally while its regional short-term contribution is weaker [35].

TABLE I
ETAS IGPE VS THE POISSON NULL, 60-DAY LEAKAGE-FREE PER-REGION BACK-ANALYSIS. SKILL CONCENTRATES IN THE HIGH-SEISMICITY, TRIGGERING-RICH REGIME.

Region	Class	IGPE (1 d)	IGPE (7 d)
Japan	high	+0.072	+0.062
Chile	high	+0.0015	+0.0024
California	high	~ 0	+0.0016
New Zealand	high	~ 0	−0.001
Central US	low	0.0	0.0
Western Europe	low	0.0	0.0
Eastern Australia	low	0.0	0.0

VI. RESULTS

A. ETAS adds skill only where there is triggering

The first question is whether ETAS only appears skilful because it over-fits high-seismicity margins. A 60-day, leakage-free, per-region back-analysis (Table I, IGPE vs the Poisson null at one and seven days) answers it: ETAS adds skill in the densest triggering regime (Japan, +0.072 at one day) and collapses to exactly the null in low-seismicity interiors (Central US, Western Europe, Eastern Australia). A single-window multi-model benchmark (Table II) tells the same story at larger absolute scale and shows that the equal-weight ensemble of heterogeneous models under-performs ETAS, because averaging in the null and the weak Reasenber–Jones dilutes the triggering signal that is the skill. Low-seismicity 0.0 is not a failure; it is the correct statement that a self-exciting model has nothing to add without sequences.

TABLE II

SINGLE-WINDOW MULTI-MODEL BENCHMARK, IGPE VS THE POISSON NULL (NATS), ON THE HELD-OUT TAIL. THE EQUAL-WEIGHT ENSEMBLE IS CONSISTENTLY WORSE THAN ETAS.

View	n_{obs}	ETAS	ensemble	R-J
Global	248	+3.712	+2.795	-0.025
Chile	11	+0.036	-0.061	-0.182
Japan	5	-0.081	-0.013	-0.438
New Zealand	2	-0.007	-0.012	-0.038

TABLE III

PRE-REGISTERED MULTI-REGION ADOPTION TEST FOR THE ETAS-FAMILY STACK, SEVEN-DAY HORIZON, 14 WEEKLY WINDOWS. THE GLOBAL GAIN IS SIGNIFICANT; REGIONAL GENERALIZATION FAILS THE RULE.

View	mean IGPE vs base	windows +	paired- t CI
Global	+0.0111	13/14	excludes zero (+)
Chile	-0.0001	3/10	crosses zero
Japan	-0.0008	8/13	crosses zero
California	+0.0077	3/3 (noise)	crosses zero
New Zealand	-0.0008	0/3	excludes zero (-)

B. ETAS-family stacking: a global gain that does not generalize

The only prospective lever the literature supports over a single well-fit ETAS is a score-weighted ETAS-family ensemble, by a small and often non-significant margin [32], [33]. We build a convex, log-score-optimal stack of three ETAS-family variants (base tiled, short-memory, long-memory), the smoothed null entering only as each member's background, weights shrunk toward the base vertex, with a pre-registered rule requiring a positive, CI-excludes-zero gain on the global view and in at least two high-seismicity regions at seven days. The 14-window back-analysis (Table III) earns a significant global gain (+0.011 over 751 events, 13/14 windows positive), carried by the long-memory variant capturing late-aftershock structure the base cutoff misses; but only 1/4 high-seismicity regions is positive (California, on three events, noise). Under the pre-registered rule the stack is **rejected**. A temporally adaptive variant appears to rescue it (3/4 regions positive) but does not withstand closer analysis: it reduces the global gain, and the regional “improvements” are sign-flipping noise on 3–54 events (no regional CI excludes zero), a multiple-comparison artifact. The regional seven-day $M \geq 5$ test is under-powered; the global view is the only adequately powered test, and there the static stack already gives the answer.

C. The short-horizon failure is over-dispersion, not shape

Why does the base ETAS N-test fail on active windows? A catalog-based number test, an ETAS branching-process forward simulation, gives the in-window count distribution including secondary triggering (Fig. 2). On a 30-day global window the frozen-intensity forecast predicts $N_{\text{fore}} = 64.3$ (background 60.1 plus first-generation triggering 4.2), the full cascade lifts the mean to 82.1 (Fano factor 1.8), and the observed 96 is consistent: the Poisson N-test fails at quantile 10^{-4} while the catalog-based test passes at 0.13. Two findings

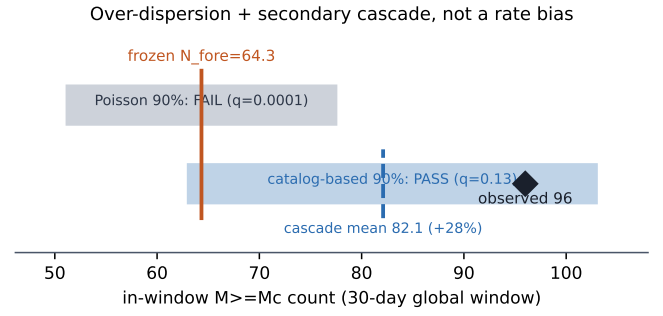


Fig. 2. Over-dispersion-aware consistency test on a 30-day global window. The frozen-intensity forecast predicts $N_{\text{fore}} = 64.3$; the full ETAS branching cascade lifts the mean to 82.1 (+28%) with a wide count distribution, so the observed 96 is consistent. The Poisson number test rejects (fail, $q = 10^{-4}$); the catalog-based test does not (pass, $q = 0.13$). Data: e13_catalog_based.json.

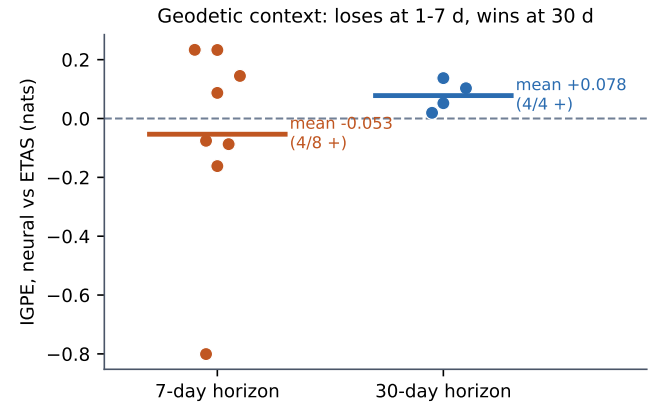


Fig. 3. The strain-conditioned neural challenger vs ETAS in the pseudo-prospective back-analysis (the exploratory dual-horizon run, fit once). Per-window IGPE is scattered around zero at seven days (mean -0.053 , only 4/8 positive) but positive in all four completed thirty-day windows (mean $+0.078$); the dedicated ten-window thirty-day validation (Fig. 4) scores $+0.115$ over 2166 events. Data: neural_pseudoprospective.json.

follow. The apparent under-forecast is over-dispersion plus secondary cascade, not a rate bias; and the frozen-intensity forecast systematically *under-counts by about 28%* because it omits within-window secondary triggering, so the catalog-based test is the correct consistency tool. This is one 30-day global window; the same test applies to any window.

D. The horizon-dependent value of geodetic context

The headline concerns the strain-conditioned neural challenger. Diagnosing it required first fixing a calibration failure: the raw model over-predicted the 30-day global rate about $490\times$ (122,148 vs 248 observed) because the training compensator constrained the background only at event locations. A fit-time rate anchor fixes it: the calibrated model forecasts 218.9 events (ETAS 173.7, observed 248) and passes the N-test. Only the calibrated model is scored below.

At the one-to-seven-day operational horizon the geodetic context does not beat ETAS (Fig. 3): an eight-window back-analysis gives mean IGPE vs ETAS -0.053 , positive in only

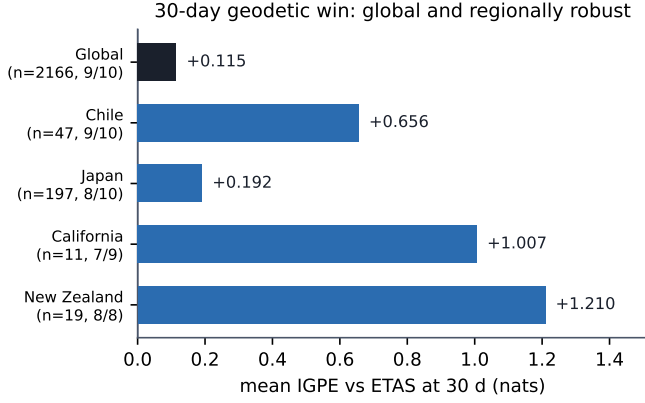


Fig. 4. Thirty-day IGPE of the geodetic-context neural over ETAS, per region (10 weekly windows). The gain is global (well-powered, $n = 2166$) and positive in every high-seismicity region. Data: outlook-evidence-30d.json.

4/8 windows, with high per-window variance (a single badly-missed sequence window drives -0.80). The ETAS-family stack of Section VI-B likewise gains only $+0.009$ to $+0.011$ globally at seven days (two back-analysis runs; Table III). The one-to-seven-day mean-rate IGPE is at or near a practical ceiling.

At thirty days the same geodetic context beats ETAS robustly and regionally (Fig. 4): a ten-window back-analysis gives a global mean IGPE vs ETAS of $+0.115$ over 2166 events, positive in 9/10 windows with a paired- t lower bound of $+0.087$, and positive in every high-seismicity region (Chile $+0.656$, Japan $+0.192$, California $+1.007$, New Zealand $+1.210$; these regional counts are small, e.g. California $n = 11$ and New Zealand $n = 19$, so the well-powered claim is the global $n = 2166$). The mechanism (Fig. 5) is explicit: the calibrated model’s total forecast is essentially constant across windows (≈ 18 at seven days, 77.3 at thirty, ratio $\approx 30/7$), so it deploys a time-flat geodetic background rather than a triggering model. That is why it loses at seven days, where ETAS triggering dominates the integral, and wins at thirty, where a better stationary background dominates it. The operational consequence is that the one-to-seven-day product remains ETAS, and the geodetic context is used only in a longer, background-dominated outlook.

VII. DISCUSSION

The four results cohere into one message (Table IV): for global $M \geq 5$ short-term forecasting, base tiled ETAS is at or near the practical ceiling on mean-rate IGPE, and the remaining levers lie elsewhere. Our review of 2024–2026 proposals (time-varying b , regime-gated neural routing, anisotropic finite-fault kernels, magnitude-head networks, multi-resolution scoring) found none supported as a large short-horizon gain, consistent with the EarthquakeNPP conclusion [9] and with our own measurements. The practical gains lie not in a larger seven-day number but in over-dispersion-aware scoring (catalog-based tests, Section VI-C), in using a covariate at the horizon where it earns skill (Section VI-D), and in rejecting re-

TABLE IV
SUMMARY OF THE FOUR FINDINGS AND THE OPERATIONAL DECISION EACH SUPPORTS UNDER THE PRE-REGISTERED RULES. EVERY VALUE IS REPRODUCIBLE FROM THE RELEASED ARTIFACTS (SECTIONS VI-A–VI-D).

Finding	Evidence (well-powered)	Decision
ETAS skill requires triggering	Japan $+0.072$ nats/1 d; low-seismicity interiors exactly 0.0	keep ETAS only where triggering exists
ETAS-family stack gains globally, not regionally	global $+0.011$ over 751 events (13/14 windows); 1/4 regions	rejected
Short-horizon N-test failure is over-dispersion	catalog test passes $q=0.13$ vs Poisson fails $q=10^{-4}$ ($+28\%$ cascade)	adopt catalog-based scoring
Geodetic covariate is horizon-dependent	7 d -0.053 ; 30 d $+0.115$ over 2166 events	adopted as 30 d outlook; 7 d remains ETAS

sults that do not generalize (Section VI-B). Methodologically, pre-registered adoption rules, shuffled-label negative controls, and the distinction between a well-powered global test (751–2166 events) and an under-powered regional test (3–54 events) are what separate a real effect from a multiple-comparison artifact; the temporally adaptive stack, which a naive reading would have adopted, is the clearest example.

VIII. LIMITATIONS

The pseudo-prospective back-analysis spans weekly windows over a limited recent period; the event counts are large (2166 global events at thirty days) but the temporal span is short, so the results are a well-powered snapshot rather than a multi-year prospective record. The neural challenger’s absolute calibration, though fixed for scoring, is not production-hardened, and it is not used in the daily one-to-seven-day product. Only one context channel (GNSS strain) is active; the remaining geophysical covariates are zero-filled and no result depends on them. The magnitude homogenization uses literature M_w relations as a documented fallback, with a bounded b -value bias the per-tile fit largely removes. The secondary-cascade under-count is measured on one 30-day global window. Each of these limitations corresponds to a flag or an artifact in the released code.

IX. CONCLUSION

We built a complete operational earthquake forecasting system and, on it, a leakage-free multi-region CSEP testbed with pre-registered evaluation, and used it to measure four things: that ETAS earns skill only where triggering exists; that an ETAS-family stack gains globally but not regionally; that the binding short-horizon consistency failure is count over-dispersion with an approximately 28% secondary-cascade under-count; and that the value of a GNSS-strain geodetic covariate is horizon-dependent, absent at one to seven days and robust at thirty, because the calibrated model contributes a time-flat background rather than triggering. The practical recommendation is to issue calibrated ETAS forecasts at the operational horizon, score with catalog-based over-dispersion

The calibrated neural deploys a time-flat geodetic BACKGROUND, not triggering (total forecast constant across windows; ratio 30/7)

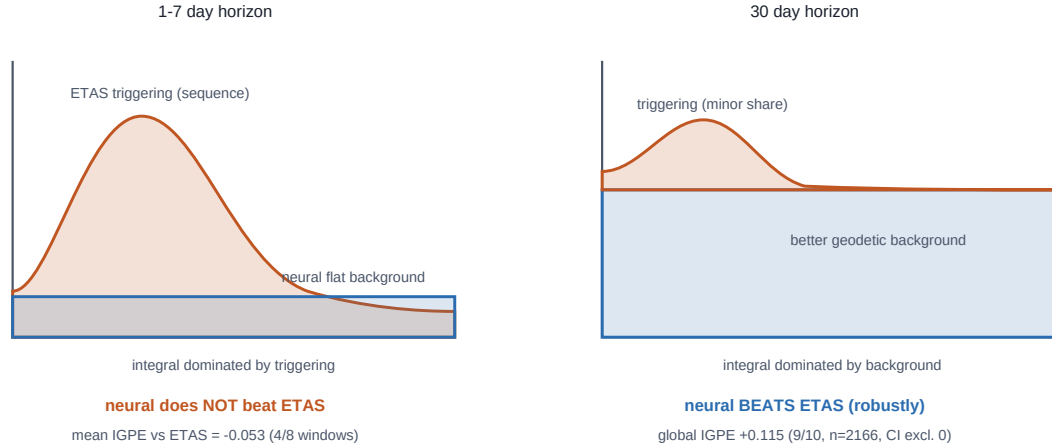


Fig. 5. Mechanism of the horizon dependence. The calibrated neural’s total forecast is constant across windows (≈ 18 events at seven days, 77.3 at thirty, ratio $\approx 30/7$): it deploys a time-flat geodetic *background*, not a triggering model. At the short horizon the integral is dominated by ETAS triggering, which the flat background cannot track (neural loses); at the long horizon the integral is dominated by the background, where a better geodetic background wins.

tests, and reserve the geodetic context for a longer background-dominated outlook. All code, configurations, provenance manifests, and artifacts are released openly, so that every number, including the negative results, is reproducible.

DATA AND CODE AVAILABILITY

Code, configurations, provenance manifests, and the daily and back-analysis artifacts, including `outlook-evidence-30d.json`, `neural_pseudoprospective.json`, `benchmark.json`, `e12_multiregion.json`, `e13_catalog_based.json`, and the per-region back-analyses, are at https://github.com/fsantibanezleal/CAOS_SEISMIC under the MIT license; a static forecast viewer is available at <https://seismic.fasl-work.com>. Catalog data are from USGS ComCat (public domain) and the Centro Sismológico Nacional, Chile; long-term anchoring from ISC-GEM; mechanisms from the Global CMT project; geodetic velocities from the Nevada Geodetic Laboratory. The figures are regenerated deterministically from these artifacts by `manuscripts/oef-geodetic-horizon/figures/make_figs.py`.

APPENDIX A

PER-REGION BACK-ANALYSIS DETAIL

The 60-day per-region back-analysis (Table I) uses the CSN, ComCat, JMA and GeoNet catalogs for Chile, California, Japan and New Zealand respectively, with region polygons, horizons, thresholds, grid and declustering pre-registered and frozen across the test split; the low-seismicity interiors (Central US, Western Europe, Eastern Australia) are included specifically to expose the high-versus-low bias. Each region $\times$ horizon cell, including the failures, is available in `results/backanalysis-<region>-* .json`.

APPENDIX B

EXPERIMENT INDEX

The results in Section VI are produced by the repository experiments E5 (per-region bias), E9 (multi-model benchmark), E10–E11 (neural challenger and pseudo-prospective horizon), E12 (ETAS-family stacking and its adaptive variant), E13 (over-dispersion and secondary cascade), and E14 (30-day geodetic validation). Each experiment records its configuration, its measured result, and the model change it motivated. The repository documentation also describes the computational methods that make the multi-region back-analysis tractable: closed-form Omori integrals, KD-tree neighbour queries, reconditioning, and vectorized neural scoring.

REFERENCES

- [1] R. J. Geller *et al.*, “Earthquakes cannot be predicted,” *Science*, vol. 275, no. 5306, p. 1616, 1997, doi:10.1126/science.275.5306.1616.
- [2] T. H. Jordan *et al.*, “Operational earthquake forecasting: State of knowledge and guidelines for utilization,” *Ann. Geophys.*, vol. 54, no. 4, 2011, doi:10.4401/ag-5350.
- [3] Y. Ogata, “Statistical models for earthquake occurrences and residual analysis for point processes,” *J. Amer. Statist. Assoc.*, vol. 83, no. 401, pp. 9–27, 1988, doi:10.1080/01621459.1988.10478560.
- [4] Y. Ogata, “Space-time point-process models for earthquake occurrences,” *Ann. Inst. Statist. Math.*, vol. 50, no. 2, pp. 379–402, 1998, doi:10.1023/A:1003403601725.
- [5] P. A. Reasenberg and L. M. Jones, “Earthquake hazard after a mainshock in California,” *Science*, vol. 243, no. 4895, pp. 1173–1176, 1989, doi:10.1126/science.243.4895.1173.
- [6] M. T. Page *et al.*, “Three ingredients for improved global aftershock forecasts,” *Bull. Seismol. Soc. Amer.*, vol. 106, no. 5, pp. 2290–2301, 2016, doi:10.1785/0120160073.
- [7] I. Spassiani *et al.*, “Operational earthquake forecasting in Italy: Validation after 10 yr of operativity,” *Geophys. J. Int.*, vol. 234, no. 3, pp. 2501–2518, 2023, doi:10.1093/gji/ggad256.
- [8] F. Serafini *et al.*, “A benchmark database of ten years of prospective next-day earthquake forecasts,” *Sci. Data*, vol. 12, p. 1501, 2025, doi:10.1038/s41597-025-05766-3.
- [9] S. Stockman, D. J. Lawson, and M. J. Werner, “EarthquakeNPP: A benchmark for earthquake forecasting with neural point processes,” *Trans. Mach. Learn. Res.*, 2026, arXiv:2410.08226.

- [10] K. Dascher-Cousineau *et al.*, “Using deep learning for flexible and scalable earthquake forecasting,” *Geophys. Res. Lett.*, vol. 50, e2023GL103909, 2023, doi:10.1029/2023GL103909.
- [11] O. Zlydenko *et al.*, “A neural encoder for earthquake rate forecasting,” *Sci. Rep.*, vol. 13, 2023, doi:10.1038/s41598-023-38033-9.
- [12] E. M. Scordilis, “Empirical global relations converting M_S and m_b to moment magnitude,” *J. Seismol.*, vol. 10, pp. 225–236, 2006, doi:10.1007/s10950-006-9012-4.
- [13] S. Wiemer and M. Wyss, “Minimum magnitude of completeness in earthquake catalogs,” *Bull. Seismol. Soc. Amer.*, vol. 90, no. 4, pp. 859–869, 2000, doi:10.1785/0119990114.
- [14] J. Woessner and S. Wiemer, “Assessing the quality of earthquake catalogues,” *Bull. Seismol. Soc. Amer.*, vol. 95, no. 2, pp. 684–698, 2005, doi:10.1785/0120040007.
- [15] Y. Shi and B. A. Bolt, “The standard error of the magnitude-frequency b value,” *Bull. Seismol. Soc. Amer.*, vol. 72, no. 5, pp. 1677–1687, 1982, doi:10.1785/BSSA0720051677.
- [16] J. Zhuang, Y. Ogata, and D. Vere-Jones, “Stochastic declustering of space–time earthquake occurrences,” *J. Amer. Statist. Assoc.*, vol. 97, no. 458, pp. 369–380, 2002, doi:10.1198/016214502760046925.
- [17] I. Zaliapin and Y. Ben-Zion, “Earthquake clusters in southern California I,” *J. Geophys. Res. Solid Earth*, vol. 118, no. 6, pp. 2847–2864, 2013, doi:10.1002/jgrb.50179.
- [18] A. Helmstetter, Y. Y. Kagan, and D. D. Jackson, “High-resolution time-independent grid-based forecast for $M \geq 5$ earthquakes in California,” *Seismol. Res. Lett.*, vol. 78, no. 1, pp. 78–86, 2007, doi:10.1785/gssrl.78.1.78.
- [19] D. L. Wells and K. J. Coppersmith, “New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement,” *Bull. Seismol. Soc. Amer.*, vol. 84, no. 4, pp. 974–1002, 1994, doi:10.1785/BSSA0840040974.
- [20] M. Schneider *et al.*, “Effective uncertainty visualization for aftershock forecast maps,” *Nat. Hazards Earth Syst. Sci.*, vol. 22, pp. 1499–1518, 2022, doi:10.5194/nhess-22-1499-2022.
- [21] Y. Y. Kagan, “Earthquake number forecasts testing,” *Geophys. J. Int.*, vol. 211, no. 1, pp. 335–345, 2017, doi:10.1093/gji/ggx300.
- [22] W. H. Savran *et al.*, “pyCSEP: A Python toolkit for earthquake forecast developers,” *Seismol. Res. Lett.*, vol. 93, no. 5, pp. 2858–2870, 2022, doi:10.1785/0220220033.
- [23] W. H. Savran *et al.*, “Pseudoprospective evaluation of UCERF3-ETAS forecasts during the 2019 Ridgecrest sequence,” *Bull. Seismol. Soc. Amer.*, vol. 110, no. 4, pp. 1799–1817, 2020, doi:10.1785/0120200026.
- [24] L. Mizrahi *et al.*, “Developing, testing, and communicating earthquake forecasts,” *Rev. Geophys.*, vol. 62, 2024, doi:10.1029/2023RG000823.
- [25] D. Schorlemmer *et al.*, “Earthquake likelihood model testing,” *Seismol. Res. Lett.*, vol. 78, no. 1, pp. 17–29, 2007, doi:10.1785/gssrl.78.1.17.
- [26] J. D. Zechar, M. C. Gerstenberger, and D. A. Rhoades, “Likelihood-based tests for evaluating space–rate–magnitude earthquake forecasts,” *Bull. Seismol. Soc. Amer.*, vol. 100, no. 3, pp. 1184–1195, 2010, doi:10.1785/0120090192.
- [27] D. A. Rhoades *et al.*, “Efficient testing of earthquake forecasting models,” *Acta Geophys.*, vol. 59, no. 4, pp. 728–747, 2011, doi:10.2478/s11600-011-0013-5.
- [28] P. M. R. DeVries *et al.*, “Deep learning of aftershock patterns following large earthquakes,” *Nature*, vol. 560, pp. 632–634, 2018, doi:10.1038/s41586-018-0438-y.
- [29] A. Mignan and M. Broccardo, “One neuron versus deep learning in aftershock prediction,” *Nature*, vol. 574, pp. E1–E3, 2019, doi:10.1038/s41586-019-1582-8.
- [30] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *J. Amer. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007, doi:10.1198/016214506000001437.
- [31] J. D. Zechar and T. H. Jordan, “Testing alarm-based earthquake predictions,” *Geophys. J. Int.*, vol. 172, no. 2, pp. 715–724, 2008, doi:10.1111/j.1365-246X.2007.03676.x.
- [32] W. Marzocchi, J. D. Zechar, and T. H. Jordan, “Bayesian forecast evaluation and ensemble earthquake forecasting,” *Bull. Seismol. Soc. Amer.*, vol. 102, no. 6, pp. 2574–2584, 2012, doi:10.1785/0120110327.
- [33] M. Herrmann and W. Marzocchi, “Maximizing the forecasting skill of an ensemble model,” *Geophys. J. Int.*, vol. 234, no. 1, pp. 73–87, 2023, doi:10.1093/gji/ggad020.
- [34] P. Bird *et al.*, “GEAR1: A global earthquake activity rate model,” *Bull. Seismol. Soc. Amer.*, vol. 105, no. 5, pp. 2538–2554, 2015, doi:10.1785/0120150058.
- [35] J. A. Bayona *et al.*, “Prospective evaluation of multiplicative hybrid earthquake forecasting models in California,” *Geophys. J. Int.*, vol. 229, no. 3, pp. 1736–1753, 2022, doi:10.1093/gji/ggac018.
- [36] G. Blewitt *et al.*, “MIDAS robust trend estimator for accurate GPS station velocities,” *J. Geophys. Res. Solid Earth*, vol. 121, no. 3, pp. 2054–2068, 2016, doi:10.1002/2015JB012552.
- [37] P. Bird, “An updated digital model of plate boundaries,” *Geochem. Geophys. Geosyst.*, vol. 4, no. 3, 1027, 2003, doi:10.1029/2001GC000252.